

MARKING SCHEME
OBJECT-ORIENTED PROGRAMMING (OOP)
(Theory Paper — 4 Questions × 5 Marks = 20 Marks)

Q.N o.	EXPECTED OUTCOMES / VALUE POINTS	Marks
SECTION – THEORY (4 × 5 Marks = 20 Marks)		
1	Definition: Inheritance is the OOP mechanism by which a derived (child) class acquires the properties and behaviours of a base (parent) class / establishes an “is-a” relationship / enables code reusability.	2
	Types of Inheritance (any three, correctly named with brief explanation): Single – one base, one derived class / Multiple – one derived class, more than one base class / Multilevel – chain of derivation across generations / Hierarchical – multiple derived classes from one base class / Hybrid – combination of two or more of the above.(Any three)	3
2	Definition of OOP: A programming paradigm based on “objects” that bundle data and behaviour, organizing design around real-world entities rather than functions and logic.	1
	Encapsulation – binding data and methods that operate on it into a single unit (class), restricting direct access to some components.	1
	Abstraction – hiding implementation detail and complexity, exposing only essential features to the user.	1
	Inheritance – a class acquiring properties and behaviours from another class, enabling reuse.	1
	Polymorphism – a function, method, or operator behaving differently depending on the object/context.	1
3	Definition: Polymorphism (“many forms”) is the ability of a function, method, or operator to behave differently depending on the object or arguments involved.	1
	Compile-time (Static) Polymorphism – achieved through function/operator overloading; resolved at compile time.	2
	Run-time (Dynamic) Polymorphism – achieved through method overriding; resolved at run time.	2
4	Definition of Class – a user-defined blueprint/template defining the properties and behaviours common to all its objects; occupies no memory on its own.	2½
	Definition of Object – a real-world instance of a class with its own actual property values; memory is allocated when it is created.	2½

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